Compiled Solutions – Senior Division

1. **01-02 C2 Bit String Flicking**

   Simplifying each term gives: RCIRC–2 11000 = 00110, 11110
   LSHIFT–1 01100 = 11000, 00110 XOR 11000 = 11110

2. **01-02 C2 Bit String Flicking**

   Let X = abcde 8
   00101 AND abcde = 00c0e so 01010 OR 00c0e = 01c1e
   RCIRC–3 01c1e = c1e01. Therefore, c1e01 = 11001 implies
   a = ∗, b = ∗, c = 1, d = ∗, e = 0. ∗∗1∗0 gives 2^3 or 8 unique solutions.

3. **02-03 C2 Bit String Flicking**

   ((RCIRC–2 01101 OR LSHIFT–1 01011) AND (NOT (LCIRC– 3 (RSHIFT–2 01110))))
   = ((01011 OR 10110) AND 00111)
   = (11111 AND 00111) = 00111 00111

4. **02-03 C2 Bit String Flicking**

   (LCIRC–3 (RSHIFT–2 (NOT 01001 AND X))) = (RCIRC–1 (LSHIFT–1 01000))
   LET X = abcde 0*1**
   (LCIRC–3 (RSHIFT–2 (NOT 01001 AND abcde))) = (RCIRC–1 (LSHIFT–1 01000))
   (LCIRC–3 (RSHIFT–2 (10110 AND abcde))) = (RCIRC–1 10000)
   (LCIRC–3 (RSHIFT–2 a0cd0)) = 01000
   (LCIRC–3 00a0c) = 01000 so 0c00a = 01000. Therefore, a=0, b=*, c=1, d=*, e=*.

5. **03-04 C2 Bit String Flicking**

   ((NOT (RCIRC-2 (LSHIFT-1 10110))) OR (LCIRC-3 01110) 11100
   AND (NOT 01011)) = ((NOT (RCIRC-2 0110)) OR 10011 AND 10100
   11100 OR 10011 AND 10100 = 11100 OR 10000 = 11100

6. **03-04 C2 Bit String Flicking**

   ((10110 AND LCIRC-2 X) OR RCIRC-2 X) = 00010 01000
   Let X = ABCDE
   (LCIRC-2 X) = CDEAB and (RCIRC-2 X) = DEABC
   10110 AND CDEAB = C0EA0 and C0EA0 OR DEABC = 00010
   0 OR E = 0 ⇒ E = 0; E OR A = 0 ⇒ A = 0; 0 OR C = 0 ⇒ C = 0;
   C OR D = 0 ⇒ D = 0; and A OR B = 1 ⇒ B = 1 so it's only 01000.

7. **04-05 C2 Bit String Flicking**

(RSHIFT-2 01101) = 00011 (10110 AND 01010) = 00010 11101
(LCIRC-3 00011) = 11000 (NOT 00010) = 11101
(LSHIFT-1 11000) = 10000 (00100 OR 11101) = 11101
(RCIRC-2 10000) = 00100

8. **04-05 C2 Bit String Flicking**

Let X = abcde 1*00*
(RCIRC–2 10110) = 10101 so (NOT 10101) = 01010
(RCIRC-2 X) = deabc so 01010 OR deabc = d1a1c
(LCIRC–3 11101) = 01111 so 01111 AND 1110 = 01110
d1a1c = 01110 implies a=1, b=*, c=0, d=0, e=*.

9. **05-06 C2 Bit String Flicking**

(RCIRC-3 (SWAP-2,2,4 (LSHIFT-2 0011011))) = 0001111
(RCIRC-3 (SWAP-2,2,4 1101100)) = (RCIRC-3 1111000) = 0001111

10. **05-06 C2 Bit String Flicking**

(RCIRC-3 (LSHIFT –1 X OR 01101)) = 01101 *0**1
Let X= abcde (LSHIFT –1 abcde) = bcde0
bcde0 OR 01101 = b11e1 (RCIRC-3 b11e1) = 1e1b1
1e1b1= 11101 $\Rightarrow$ b=0, c=∗, d=∗, e=1, a=∗ so abcde = *0**1.

11. **06-07 C2 Bit String Flicking**

A26 = 1 0 1 0 0 0 1 0 0 1 1 0 D93
7B5 = 0 1 1 1 1 0 1 1 0 1 0 1
XOR=1 1 0 1 1 0 0 1 0 0 1 1 = D93

12. **06-07 C2 Bit String Flicking**

((LCIRC-3 (X AND 10010)) OR (RSHIFT-2 (X AND 01101)) = 10011 0111*
Let X = a b c d e a b c d e
AND 1 0 0 1 0 AND 0 1 1 0 1
a 0 0 d 0 0 b c 0 e
(LCIRC-3 a 0 0 d 0) = d 0 a 0 0 (RSHIFT-2 0 b c 0 e) = 0 0 0 b c
(d 0 a 0 0 OR 0 0 0 b c) = d 0 a b c = 1 0 0 11
a = 0, b = 1, c = 1, d = 1, e = *

13. **07-08 C2 Bit String Flicking**

(RCIRC-3 01101 OR (LSHIFT-2 (RCIRC-1 01110) AND 10111)) 10101
= (10101 OR (LSHIFT-2 00111 AND 10111))
= (10101 OR (11100 AND 10111)) = (10101 OR 10100) = 10101

14. **07-08 C2 Bit String Flicking**

((LCIRC-2 (RSHIFT-1 NOT X)) AND (RCIRC-1 (LCIRC-3 01101))) 01*0*
Let X = abcde
(LCIRC-2 (RSHIFT-1 NOT abcde)) = (LCIRC-2 (RSHIFT-1 ABCDE))
= (LCIRC-2 0ABCD) = BCD0A
(RCIRC-1 (LCIRC-3 01101)) = (RCIRC-1 01011) = 10101
BCD0A AND 10101 = 00101 $\Rightarrow$ B=0,C=*,D=1,A=1,E=*
$\Rightarrow$ a=0, b=1, c=*, d=0, e=*

15. **08-09 C2 Bit String Flicking**

First convert to binary: 3547 or 3547_8
110111101011 XOR 101011001110 AND 101110101101
AND is evaluated before XOR yielding 011 101 100 $111_2 = 3547_8$

16. **08-09 C2 Bit String Flicking**

:

(RCIRC-2 (RSHIFT-1 X)) = (RSHIFT-1 10101) OR (LCIRC-2 01010) 1101*
Let X = abcde
RHS = (RSHIFT-1 10101) OR (LCIRC-2 01010)
= 01010 OR 01001 = 01011
LHS = (RCIRC-2 (RSHIFT-1 abcde)) =(RCIRC-2 0abcd) = cd0ab
LHS = RHS $\Rightarrow$ cd0ab = 01011 $\Rightarrow$ a=1, b=1, c=0, d=1, e=*
Therefore 1101* is the solution.

17. **09-10 C2 Bit String Flicking**

(LCIRC-2 (RSHIFT-2 01101)) = (LCIRC-2 00011) = 01100 01111
(NOT 10110 AND RCIRC-3 10110) = 01001 AND 11010 = 01000
(RSHIFT-2 (LCIRC-3 NOT 01100)) = (RSHIFT-2 (LCIRC-3 10011))
= (RSHIFT-2 11100) = 00111
01100 OR 01000 XOR 00111 = 01100 OR (01000 XOR 00111)
= 01100 OR 01111 = 01111

18. **09-10 C2 Bit String Flicking**

Let X = abcde and NOT X = ABCDE 6
(RCIRC-3 (LSHIFT-2 (NOT X))) = (RCIRC-3 (LSHIFT-2 ABCDE))
= (RCIRC-3 CDE00) = E00CD
(RSHIFT-1 (LCIRC-2 X)) = (RSHIFT-1 (LCIRC-2 abcde))
= (RSHIFT-1 cdeab) = 0cdea
So E00CD AND 0cdea = 00001⇒ b = *, a = 1, d = 0.
Then if e = 1 then c = 1, but if e = 0 then c = *.
∴ There are 6 solutions: 1110*, 1*000, 1010*.

19. **10-11 C2 Bit String Flicking**

(LCIRC-2 11011) = 01111, 01111
(NOT 11010) = 00101, (LCIRC-2 01100) = 10001
So 01111 OR 00101 AND (RSHIFT-1 (10001 AND 10101))
= 01111 OR 11011 AND (RSHIFT-1 10001)
= 01111 OR 00101 AND 01000 = 01111 OR 00000 = 01111

20. **10-11 C2 Bit String Flicking**

Let X = abcde 0*101
(RCIRC-3 X) = (RCIRC-3 abcde) = cdeab
(NOT cdeab) = CDEAB
(RSHIFT-1 CDEAB) = 0CDEA
(LCIRC-2 0CDEA = DEA0C
DEA0C = 10100 ⇒ D=1, E=0, A=1, C=0
∴a=0, b=*, c=1, d=0, e=1

21. **11-12 C2 Bit String Flicking**

(LCIRC−2 01110) = 11001 11100
(RSHIFT−1 01011) = 00101
XOR 11100

22. **11-12 C2 Bit String Flicking**

Let X = abcde 1**01
(RSHIFT−1 (LCIRC−2 abcde)) = (RSHIFT−1 cdeab) = 0cdea
(NOT 10111) = 01000
OR 01dea
01dea = 01011 ⇒ a = 1, b = *, c = *, d = 0, e = 1
∴ abcde = 1**01

23. **11-12 C3 Bit String Flicking**

Let X = abcde 1*111
LHS = (RSHIFT-1 (LCIRC-2 X)) = (RSHIFT-1 (LCIRC-2 abcde))
= (RSHIFT-1 cdeab) = 0cdea
RHS = (RCIRC-2 01101) OR (LSHIFT-2 10101) AND (NOT 11001)
= 01011 OR (10100 AND 00110) = 01011 OR 00100 = 01111
0cdea = 01111 ⇒ a = 1, b = *, c = 1, d = 1, e = 1 ∴1*111

24. **12-13 C2 Bit String Flicking**

(LCIRC-2 (RSHIFT-1 (NOT 01110))) 00001
= (LCIRC-2 (RSHIFT-1 10001))
= (LCIRC-2 01000)
= 00001

25. **12-13 C2 Bit String Flicking**

((RCIRC-2 (01010 XOR 10010)) OR (LSHIFT-1 (NOT 01110))) 00110
= ((RCIRC-2 (01010 XOR 10010)) OR (LSHIFT-1 10001))
= (RCIRC-2 11000) OR 00010
= 00110 OR 00010
= 00110

26. **12-13 C3 Bit String Flicking**

Let X = abcde 0**0*
LHS = (LCIRC-2 01011) OR X
= 01101 OR abcde = a11d1
RHS = RSHIFT-1 11011 = 01101
a11d1 = 01101→ a = 0, b = *, c = *, d = 0, e = * → 0**0*

27. **13-14 C2 Bit String Flicking**

(RSHIFT-2 (LCIRC-1 01011)) OR (NOT (LSHIFT-1 (RCIRC-3 10110))) 01111
= (RSHIFT-2 10110) OR (NOT (LSHIFT-1 11010))
= 00101 OR (NOT 10100) = 00101 OR 01011 = 01111

28. **13-14 C2 Bit String Flicking**

(RCIRC-2 (LSHIFT-1 X)) = 10111
Let X = abcde
LHS = ((RCIRC-2 (LSHIFT-1 X))
= ((RCIRC-2 (LSHIFT-1 abcde))
= (RCIRC-2 bcde0) =e0bcd
e0bcd = 10111 → e = 1, b = 1, c = 1, d = 1, a = * → *1111

29. **13-14 C3 Bit String Flicking**

Let X = abcde
LHS = (LSHIFT-2 (RCIRC-1 01110)) OR (LCIRC-2 (RSHIFT-1 abcde)) 1****
AND (NOT (RCIRC-1 (LSHIFT-2 01010))
= (LSHIFT-2 00111) OR (LCIRC-2 0abcd) AND (NOT (RCIRC-1 01000)
= 11100 OR bcd0a AND (NOT 00100) = 11100 OR bcd0a AND 11011 =
11100 OR bc00a = 1110a
RHS = (LCIRC-2 (NOT 10000)) = (LCIRC-2 01111) = 11101
1110a = 11101→ a = 1, b = *, c = *, d = *, e = * → ****

30. **14-15 C2 Bit String Flicking**

(LCIRC-2 (RSHIFT-1 (NOT 01110))) 11000
= (LCIRC-2 (RSHIFT-1 10001))
= (LCIRC-2 01000)
= 00001

31. **14-15 C2 Bit String Flicking**

((RCIRC-2 (01010 XOR 10010)) OR (LSHIFT-1 (NOT 01110))) 2
= ((RCIRC-2 (01010 XOR 10010)) OR (LSHIFT-1 10001))
= (RCIRC-2 11000) OR 00010
= 00110 OR 00010
= 00110

32. **14-15 C3 Bit String Flicking**

Let X = abcde 1110*, 1000*
LHS = (NOT ((LCIRC-2 01011) AND (RSHIFT-1 X)))
= (NOT (01101 AND 0abcd))
= (NOT 0ab0d) = 1AB1D
RHS = (RCIRC-2 01110 OR 10110 AND NOT X)
= 10011 OR 10110 AND ABCDE
= 10011 OR A0CD0 = 10C11
So 1AB1D = 10C11→ A = 0, B = C , D = 1, e = * → 1110*, 1000*

33. **15-16 C2 Bit String Flicking**

(LCIRC-2 10011) OR (RSHIFT-2 01110) AND 01111
(RCIRC-2 (LSHIFT-1 01010))
= 01110 OR 00011 AND (RCIRC-2 10100)
= 01110 OR (00011 AND 00101)
= 01110 OR 00001
= 01111

34. **15-16 C2 Bit String Flicking**

Let X = abcde 00**1
LHS = (LCIRC-2 10001) OR X
= 00110 OR abcde = ab11e
RHS = (RSHIFT-1 01110)
= 00111
If ab11e = 00111, then a = 0, b = 0, c = *, d = *, e = 1
So X = 00**1

35. **15-16 C3 Bit String Flicking**

Let X = abcde ***00
LHS = ((RSHIFT-1 (LCIRC-2 (LSHIFT-1 abcde)))
OR (NOT 01110 AND (RCIRC-2 01101)))
= ((RSHIFT-1 (LCIRC-2 bcde0)) OR (10001 AND 01011))
= ((RSHIFT-1 de0bc) OR 00001)
= (0de0b OR 00001) = 0de01
RHS = (LCIRC-1 (LSHIFT-2 00100))
= (LCIRC-1 10000) = 00001
So 0de01 = 00001 → a = *, b = * , c = *, d = 0, e = 0 → ***00